

2-1 (a)	X	Y	Z	$\overline{XYZ}$	$\overline{X+Y+Z}$
	0	0	0	1	1
	0	0	1	1	1
	0	1	0	1	1
	0	1	1	1	1
	1	0	0	1	1
	1	0	1	1	1
	1	1	0	1	1
	1	1	1	0	0

$$\begin{aligned}
 2-2 (a) \quad \overline{X}\overline{Y} + \overline{X}Y + XY &= \overline{X}\overline{Y} + \overline{X}Y + \overline{X}Y + XY \\
 &= \overline{X}(Y + \overline{Y}) + Y(X + \overline{X}) \\
 &= \overline{X} + Y
 \end{aligned}$$

$$2-2 (c) \quad Y + \overline{X}Z + XY = X + Y + Z \quad ? \quad Y=0, \quad X=Z=1$$

$$\begin{aligned}
 2-3 (a) \quad ABC\overline{C} + B\overline{C}D + BC + \overline{C}D &= A\overline{B}\overline{C} + B\overline{C} + B\overline{D} + BC + \overline{C}D \\
 &= B(1 + \overline{A}\overline{C} + \overline{D}) + \overline{C}D \\
 &= B + \overline{C}D
 \end{aligned}$$

$$\begin{aligned}
 2-3 (c) \quad (\overline{A} + \overline{B} + \overline{C} + \overline{D})(A + B + C + D) \\
 &= \overline{A}A + \overline{A}B + \overline{A}C + \overline{A}D + \overline{B}A + \overline{B}B + \overline{B}C + \overline{B}D \\
 &\quad + \overline{C}A + \overline{C}B + \overline{C}C + \overline{C}D + \overline{D}A + \overline{D}B + \overline{D}C + \overline{D}D \\
 (X\overline{X}=0) \quad &= \overline{A}B + \overline{A}C + \overline{A}D + \overline{B}A + \overline{B}C + \overline{B}D + \overline{C}A + \overline{C}B + \overline{C}D + \overline{D}A + \overline{D}B + \overline{D}C \\
 (共设律) \quad &= \overline{A}D + \overline{A}B + \overline{C}D + \overline{B}C
 \end{aligned}$$

$$\begin{aligned}
 2-6 (b) \quad \overline{(A+B+C)} \cdot \overline{ABC} &= \overline{A}\overline{B}\overline{C} \cdot (\overline{A} + \overline{B} + \overline{C}) \\
 &= \overline{A}\overline{B}\overline{C}
 \end{aligned}$$

$$\begin{aligned}
 2-6(d) \quad \bar{A}\bar{B}D + \bar{A}\bar{C}D + BD &= D(\bar{A}\bar{B} + \bar{A}\bar{C} + B) \\
 &= D(\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{A}B + AB) \\
 &= D(\bar{A} + AB)
 \end{aligned}$$

2-10 (a)	X	Y	Z	$(XY+Z)(Y+XZ)$
	0	0	0	0
	0	0	1	0
	0	1	0	0
	0	1	1	1
	1	0	0	0
	1	0	1	1
	1	1	0	1
	1	1	1	1

$$\Rightarrow (XY+Z)(Y+XZ) = \sum m(3,3,6,7) = \prod M(0,1,2,4)$$

2-11 (a) E 的最小项: $\bar{x} + \bar{y} + \bar{z}, x + \bar{y} + \bar{z},$
 $\bar{x} + y + \bar{z}, x + y + \bar{z}$
 F 的最小项: $\bar{x}\bar{y}z, \bar{x}y\bar{z}, x\bar{y}\bar{z}, xy\bar{z}.$

F 的最小项: $x + y + \bar{z}, x + \bar{y} + \bar{z},$
 $\bar{x} + y + \bar{z}, \bar{x} + \bar{y} + \bar{z}.$
 F 的最小项: $\bar{x}\bar{y}\bar{z}, \bar{x}y\bar{z}, x\bar{y}\bar{z}, xy\bar{z}.$

2-11 (c) E+F 的最小项: $\bar{x}\bar{y}\bar{z}, \bar{x}\bar{y}z, \bar{x}y\bar{z}, x\bar{y}\bar{z},$
 $xy\bar{z}, xyz$

E·F 的最小项: $\bar{x}y\bar{z}, x\bar{y}\bar{z}$

12-10 (C)	X	Y	Z	W	$Wx\bar{Y} + Wxz + Wx\bar{Z} + \bar{Y}\bar{Z}$
	0	0	0	0	0
	0	0	0	1	0
	0	0	1	0	0
	0	0	1	1	0
	0	1	0	0	1
	0	1	0	1	1
	0	1	1	0	0
	0	1	1	1	0
	1	0	0	0	0
	1	0	0	1	1
	1	0	1	0	0
	1	0	1	1	1
	1	1	0	0	1
	1	1	0	1	1
	1	1	1	0	0
	1	1	1	1	1

$$\Rightarrow B_1 A' = \sum m(4, 5, 9, 11, 12, 13, 15)$$

$$= \prod M(0, 1, 2, 3, 6, 7, 8, 10, 14)$$

$$2-11(d) \quad E = \sum m(1, 2, 4, 6)$$

$$F = \sum m(0, 2, 4, 7)$$

2-12 (b) $\bar{x} + x(x + \bar{y})(y + \bar{z})$
 $= \bar{x} + xy + x\bar{z} = \sum m(0, 1, 2, 3, 4, 6, 7)$
 $= \pi M(5)$

2-15 (a) $\begin{array}{c|cccc} x \backslash yz & 00 & 01 & 11 & 10 \\ \hline 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{array}$ $\hat{f}(x,y,z) = xy + \bar{x}\bar{z}$

(b) $\begin{array}{c|cccc} A \backslash BC & 00 & 01 & 11 & 10 \\ \hline 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \end{array}$ $\hat{f}(A,B,C) = \bar{A} + A\bar{B}C$

(c) $\begin{array}{c|cccc} A \backslash BC & 00 & 01 & 11 & 10 \\ \hline 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \end{array}$ $\hat{f}(A,B,C) = \bar{B} + B\bar{C}$

2-17 (a) $\begin{array}{c|cccc} w \backslash x \backslash yz & 00 & 01 & 11 & 10 \\ \hline 00 & 1 & 1 & & 1 \\ 01 & 1 & & 1 & \\ 11 & 1 & & & \\ 10 & 1 & & & 1 \end{array}$ $\hat{f}(w,x,y,z) = \bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{w}x\bar{y} + \bar{w}xy\bar{z}$

(b) $\begin{array}{c|cccc} AB \backslash CD & 00 & 01 & 11 & 10 \\ \hline 00 & & 1 & & \\ 01 & 1 & 1 & & 1 \\ 11 & 1 & 1 & 1 & \\ 10 & & & 1 & 1 \end{array}$ $\hat{f}(A,B,C,D) = B\bar{C} + \bar{A}\bar{C}D + ACD + A\bar{B}C + \bar{A}B\bar{D}$

2-19 (a)

$wx \backslash yz$	00	01	11	10
00	1			1
01		1	1	
11	1	1	1	1
10	1			1

主项各取 $xz, \bar{x}\bar{z}, wx, w\bar{z}$.
均是必要的

2-22 (a)

$AB \backslash CD$	00	01	11	10
00	0	1	1	0
01	0	0	1	0
11	1	1	1	0
10	1	1	1	0

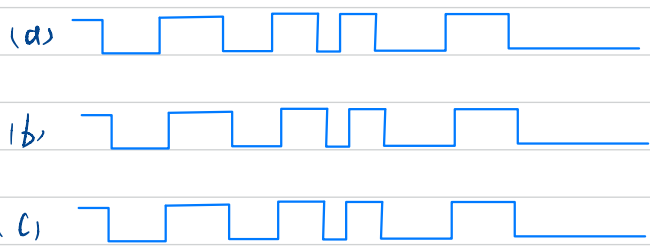
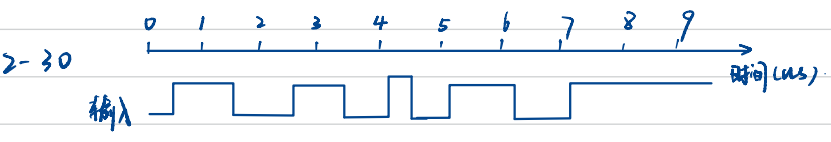
$$\begin{aligned} \bar{B}C &= A\bar{C} + CD + \bar{A}BD \\ &= (\bar{C} + D)(A + \bar{B} + C)(\bar{A} + \bar{C} + \bar{D}) \end{aligned}$$

2-25 (b)

$wx \backslash yz$	00	01	11	10
00	1			1
01	1	1	X	
11		X	1	1
10	1			X

右图中所有框出的是主项各取，其中红色框出的是
原主项各取。
化简得 $F = \bar{x}\bar{z} + xz + w\bar{x}\bar{y} + wxy$

2-29 $t_{max} = 3 \times 0.073 + 0.048 = 0.267 \text{ ns}$



2-3). (a) 从 C、D 到 F. $t_{HL1} = 4t_{pHL} = 0.80 \text{ ns}$. $t_{LH1} = 4t_{pLH} = 1.44 \text{ ns} \Rightarrow t_{pd1} = 1.12 \text{ ns}$
 从 $\bar{B}$ 到 F. $t_{HL2} = 3t_{pHL} = 0.60 \text{ ns}$. $t_{LH2} = 3t_{pLH} = 1.08 \text{ ns} \Rightarrow t_{pd2} = 0.84 \text{ ns}$
 从 A、B、 $\bar{C}$ 到 F. $t_{HL3} = 2t_{pHL} = 0.4 \text{ ns}$. $t_{LH3} = 2t_{pLH} = 0.72 \text{ ns} \Rightarrow t_{pd3} = 0.56 \text{ ns}$.

(b) 从 C、D 到 F. $t_{pd4} = 4t_{pd} = 1.12 \text{ ns}$
 从 $\bar{B}$ 到 F. $t_{pd5} = 3t_{pd} = 0.84 \text{ ns}$
 从 A、B、 $\bar{C}$ 到 F. $t_{pd6} = 2t_{pd} = 0.56 \text{ ns}$

(c) (a) (b) 结果相同. 表达的意义也没有区别